

Data-Driven Brand & Product Strategy

An R-based Analysis of YouTube and Google Trends

Daniel Huber & Dongyuan Gao

2025-09-16

Table of contents

1	Our Story	3
1.1	The Beginning of an Idea	3
1.2	How to Use This Book	3
1.2.1	Navigating through the report	3
1.2.2	This report is organized into the following chapters:	3
2	Prepare the Data for Analysis	4
2.1	Data Description	4
2.1.1	YouTube Trending Video Dataset	4
2.1.2	Google Search Interest Dataset	4
2.2	Data Preparation and Cleaning	5
3	Analysis and Visualisation	6
3.1	The Big Picture - Comparing the Brands	6
3.1.1	Graph 1: Overall Google Search Interest Comparison	6
3.1.2	Graph 2: Overall YouTube Views Comparison	7
3.1.3	Graph 3: The iPhone Hype Cycle	9
3.2	Google Search Trend Analysis	11
3.2.1	Loading and preparing the data	11
3.2.2	Correlations between brands	11
3.2.3	Seasonality of Google searches	11
3.3	Model Analysis: fitting a linear regression model and visualize it	12
3.3.1	Data visualization for 5 countries	12
3.3.2	Prepare the data for modelling and checking all correlations	13
3.3.3	Fitting a linear model	14
3.3.4	Visualize the linear regression and the quality of the model	15
3.3.5	Conclusion for our clients	15
3.4	YouTube Video Analysis	15
3.4.1	Positive YouTube engagement	16
3.4.2	Negative YouTube engagement	16
3.4.3	Longevity of YouTube videos	17
3.5	WordCloud	18
3.5.1	Rolex wordcloud	19
3.5.2	iPhone wordcloud	20
4	Chapter of choice: The R Package <i>gtrendsR</i>	22
4.1	Core Function	22
4.2	Main Parameters	22
4.3	Structure of the Results	22
4.4	Summary	23
4.5	Interpretating Google Trend Data	23
4.6	Example: World map with Google Trends data for Tesla	23
4.6.1	Import libraries	23
4.6.2	Load data	23
4.6.3	Prepare the <code>interest_by_country</code> data in <code>gtrends</code>	23
4.6.4	Load and prepare the map data using <code>ne_countries</code> function from the <code>rnatuarearth</code> package.	24
4.6.5	Producing the world map with <code>geom_sf</code>	24
5	Generative AI Usage and Guidelines	25
5.1	Guidelines for Responsible and Effective Usage of Gen-AI	25
5.2	Use cases of AI tools	25
5.3	Benefits and challenges in using Generative AI tools	25
6	Conclusion, limitations and outlook	26
6.1	Conclusions	26
6.1.1	Key Takeaways for Brands	26
6.1.2	Key takeaways for R-Bootcamp Learning	26
6.2	Limitations	26
6.3	Outlook	27

1 Our Story

Welcome to our **HSLU R Bootcamp Project**.

This project explores **Brand and Product Strategy for iPhone, Tesla and Rolex**.

By analyzing **YouTube** Trending Video and **Google** Trend data, we have conducted data analysis, visualization, and modeling analysis in R.

The objective is to demonstrate insights for hypothetical clients and to deepen our learning in R, as part of our bootcamp learning journey.

1.1 The Beginning of an Idea

Today, both YouTube and Google are central media and information sources globally, and they are key platforms for online **behavior research** and big data **insights generation**. Users on these platforms are highly engaged. This engagement provides consumer brands an opportunity to understand the behavior and preferences of customers. Analyzing data from these platforms allows the brands to provide appealing products that adhere to the needs of the market. This approach can not only increase **profitability** but also improves **efficiency** by avoiding unwanted products or advertisement campaigns.

These factors have inspired us to create a service to our hypothetical clients, to help them analyze data and explore insights. To showcase our capabilities, we analyzed YouTube and Google data from 2017 to 2018 for three hypothetical clients: iPhone, Tesla, and Rolex. This analysis serves as an example to demonstrate our methodology, potential results, and the value our service for future clients.

1.2 How to Use This Book

1.2.1 Navigating through the report

You can interact with the report in two primary ways. For dynamic experience, we recommend viewing the **HTML** version. Simply open the `**_book/index.html**` file in your web browser after rendering.

If you prefer a static document, the **PDF** version as `**_book/Data_Driven_Brand_Product_Strategy_Gao_Huber.pdf**` can be also a good choice.

Alternatively, you can play around with the code and generate new reports. However, please ensure you have R, RStudio, and Quarto installed, then in your Rstudio terminal, execute the command: `quarto render`. This command will first render the book, then both versions will be available in the `_book` folder.

1.2.2 This report is organized into the following chapters:

- **Chapter 1: Our Story:** You are right here, reading about how the idea was generated and how you should navigate through our project in chapters.
- **Chapter 2: Prepare the Data for Analysis:** Details about the datasets, the process of data extraction, cleaning, and transformation.
- **Chapter 3: Analysis and Visualization:** Presentation of the **core-findings**, from broad descriptive statistics to in-depth correlation and trend analysis. Let's understand the data with visualizations.
- **Chapter 4: Chapter of Choice - The R Package gtrendsR:** An exploration of the `gtrendsR` package, where Google trend data is extracted. We will showcase its capabilities and **technical implementation**.
- **Chapter 5: Generative AI Usage and Guidelines:** Explanation of the approach that we use to apply Generative AI in our research, including why it helps the research process and how it's implemented in a **responsible** way.
- **Chapter 6: Conclusion, Limitations, and Outlook:** A summary of the key takeaways from our research, a discussion of the limitations of the study, and potential improvement aspects for future projects.

2 Prepare the Data for Analysis

2.1 Data Description

Before explaining the data preparation process, it is important to first understand the data, including its characteristics, timeframe and quality. This section provides an **overview** of the **2 datasets** extracted for this analysis using an R package.

2.1.1 YouTube Trending Video Dataset

The first source of data is the **YouTube Trending Video Dataset** from Kaggle: * **Source:** [YouTube Trending Video Dataset on Kaggle](#). This dataset includes several months of data on daily trending YouTube videos. Data is included for many countries like the USA, UK, Mexico or Japan, with up to 200 listed trending videos per day.

- **Key Variables (Raw):** Each region's data is in a separate file. Data includes the video title, channel title, publish time, tags, views, likes and dislikes, description, and comment count.
- **Data Volume:** The data spans the period from **2017-11-14 to 2018-06-14**, which defines the timeframe for this case study. According to the source, the original data was extracted using YouTube API. Data of 5 countries is selected for our study: the US, Japan, Mexico, Germany and France. After combining the 5 countries' files into one dataframe, we got 183,487 rows and 17 columns. Furthermore, after separating the multi-valued tags(keyword) for each videos into single rows, the dataframe has 3,002,056 rows and 17 columns.
- **Initial Observations/Challenges:** The first challenge that we looked into was the `tags` column, which are the **topics/keywords** that we plan to filter and analyze. However there are many **tags** in each row, which means we will need to separate the individual tags using a delimiter.

video_id	trending_date	title	channel_title	category_id	publish_time	tags
2kyS6svSYSE	17.14.11	WE WANT TO TALK ABOUT OUR MARRIAGE	CaseyNeistat	22	2017-11-13T17:13:01.000Z	SHANteli martin
1ZAPwrtAFY	17.14.11	The Trump Presidency: Last Week Tonight with John Oliver (HBO)	LastWeekTonight	24	2017-11-13T07:30:00.000Z	last week tonight trump presidency last week tonight donald trump john ol
5qpjK5DgCt4	17.14.11	Racist Superman Rudy Mancuso, King Bach & Lele Pons	Rudy Mancuso	23	2017-11-12T19:05:24.000Z	racist superman rudy mancuso king bach racist superman love
puqaWtEC7iY	17.14.11	Nickelback Lyrics: Real or Fake?	Good Mythical Morning	24	2017-11-13T11:00:04.000Z	rhett and link gmm good mythical morning rhett and link good mythical r
d380meDOWOM	17.14.11	I Dare You: GOING BALDI?	rigahiga	24	2017-11-12T18:01:41.000Z	ryan higa higatv nigahiga i dare you idy rhpc daree no truth cc
gHZ1Qx0KIKM	17.14.11	2 Weeks with iPhone X	iJustine	28	2017-11-13T19:07:23.000Z	justine week with iPhone X iphone x apple iphone iphone x review
39idVpFF7NQ	17.14.11	Roy Moore & Jeff Sessions Cold Open - SNL	Saturday Night Live	24	2017-11-12T05:37:17.000Z	SNL Saturday Night Live SNL Season 43 Episode 1730 Tiffany Haddis
nc99ccSXST0	17.14.11	5 Ice Cream Gadgets put to the Test	CrazyRussianHacker	28	2017-11-12T21:50:37.000Z	5 Ice Cream Gadgets Ice Cream Cream Sandwich Maker gadgets gad
jr9QIXwC9vc	17.14.11	The Greatest Showman Official Trailer 2 [HD] 20th Century FOX	20th Century Fox	1	2017-11-13T14:00:23.000Z	Trailer Hugh Jackman Michelle Williams Zac Efron Zendaya Rebecca
TUmyygmMGA	17.14.11	Why the rise of the robots won't mean the end of work	Vox	25	2017-11-13T13:45:16.000Z	vox.com vox explain shift change future of work automation robots
9wRQIJFNDW8	17.14.11	Dion Lewis' 103-Yd Kick Return TD vs. Denver Can't-Miss Play NFL Wk 10 Highlights	NFL	17	2017-11-13T02:05:26.000Z	NFL Football offense defense afc nfc American Football highlight
VIQJH6A0	17.14.11	(SPOILERS) 'Shiva Saves the Day' Talked About Scene Ep. 804 The Walking Dead	amc	24	2017-11-13T03:00:00.000Z	The Walking Dead shiva tiger king ezekiel episode 804 episode 4 sr
5E4ZBSInQUU	17.14.11	Marshmello - Blocks (Official Music Video)	marshmello	10	2017-11-13T17:00:00.000Z	marshmello blocks marshmello blocks blocks music video marshmello

Figure 2.1: A screenshot of one of the raw data file

2.1.2 Google Search Interest Dataset

We obtained this dataset by **extracting** Google search data ourselves, instead of using ready-to-use data.

- **Source:** Google Trends, accessed via the `gtrendsR` R package.
- **Key Variables (from `gtrendsR` output):**
 - `date`: Date of search interest.
 - `hits`: A Google defined index of search interest (0-100).
 - `keyword`: The search term queried: iPhone, Rolex and Tesla.
 - `geo`: Geographic region of the search (we only define 5 countries that we chose).
- **Initial Observations:** After reading into the package information and data, we found out that: the hits(0-100) reflect relative search interest, not absolute search volume.

2.2 Data Preparation and Cleaning

The data preparation and cleaning was done in an exploratory way, step by step, heading towards one goal: to have a data frame ready for merging with the Google dataset. To achieve that, we have then written down the exact data structure from both sides, which contain the exact information we want. To program that, the following was needed: group data, add and delete columns, filter and merge, etc.

In summary, our primary objective was to create a **unified dataset** that captures daily trending patterns for our **target keywords** (iPhone, Tesla, and Rolex) across five countries.

Following up, we will showcase our **YouTube Data Cleaning and Preprocessing**: - Combining Country-Specific Datasets: The initial data extraction have five separate country files. Our first critical step is to read and merge them into one **single data frame**. Therefore, the country codes "JP" "US" "DE" "FR" "MX" were added to the merged table. The resulting dataset contained 183,487 observations spanning from November 2017-11-14 to 2018-06-14 across all five countries.

- **Extracting and Preparing Keywords:** The `tags` column presented both a challenge and an opportunity. We found out that each video contained multiple tags separated by pipe characters. To solve the problem, we separated then the keyword(tags) for each videos as one individual row, using syntax like:

```
#### Filter for the three keywords and create a new 'keyword' column
relevant_videos <- tags_per_day %>%
  filter(grepl("iphone|tesla|rolex", tags, ignore.case = TRUE)) %>%
  mutate(keyword = case_when(
    grepl("iphone", tags, ignore.case = TRUE) ~ "iPhone",
    grepl("tesla", tags, ignore.case = TRUE) ~ "Tesla",
    grepl("rolex", tags, ignore.case = TRUE) ~ "Rolex",
    TRUE ~ "Other"
  ))
```

Then why is it an opportunity? Actually after separating the desired iPhone, Tesla and Rolex tags, we could even explore what tags are around these products. New types of graphs like word clouds could be made.

- **Engagement Metric Calculation:** To measure audience engagement at a keyword level, new metrics were created.

To do that, firstly we aggregated the data and calculate two engagement ratios: `total_likes/total_views` and `total_comments/total_views`. These ratios provide insights into audience engagement (engagement quality) independent of video popularity. This valuable information helps companies to identify topics that can stimulate most engagement, and developing an efficient advertising strategy.

- **Final dataset Integration:** The final chapter of the data preparation phase is the creation of a single, comprehensive dataset, combining the YouTube data structure and the Google search data.
- **Merging Strategy:** The processed YouTube and Google Trend dataset were combined using a `full_join` (including trending dates, video metrics, and extracted tags, calculated engagement ratios, keyword, country) with the Google Trends search interest data. In case on a day, there is Google trend data for a keyword, but no trending video on YouTube, the metrics will be value 0. An extra column is also added, marking if the day has trending video or not. This method ensures that all data points from both sources are retained, **preventing any loss of information**.
- **Join Keys:** The join should be performed on common identifiers such as keyword/tag and date.
- **Resulting dataset:** the structure of the final merged data is

[1] "trending_date" "country" "keyword" "total_views"

[5] "total_likes" "total_dislikes" "total_comments" "video_count"

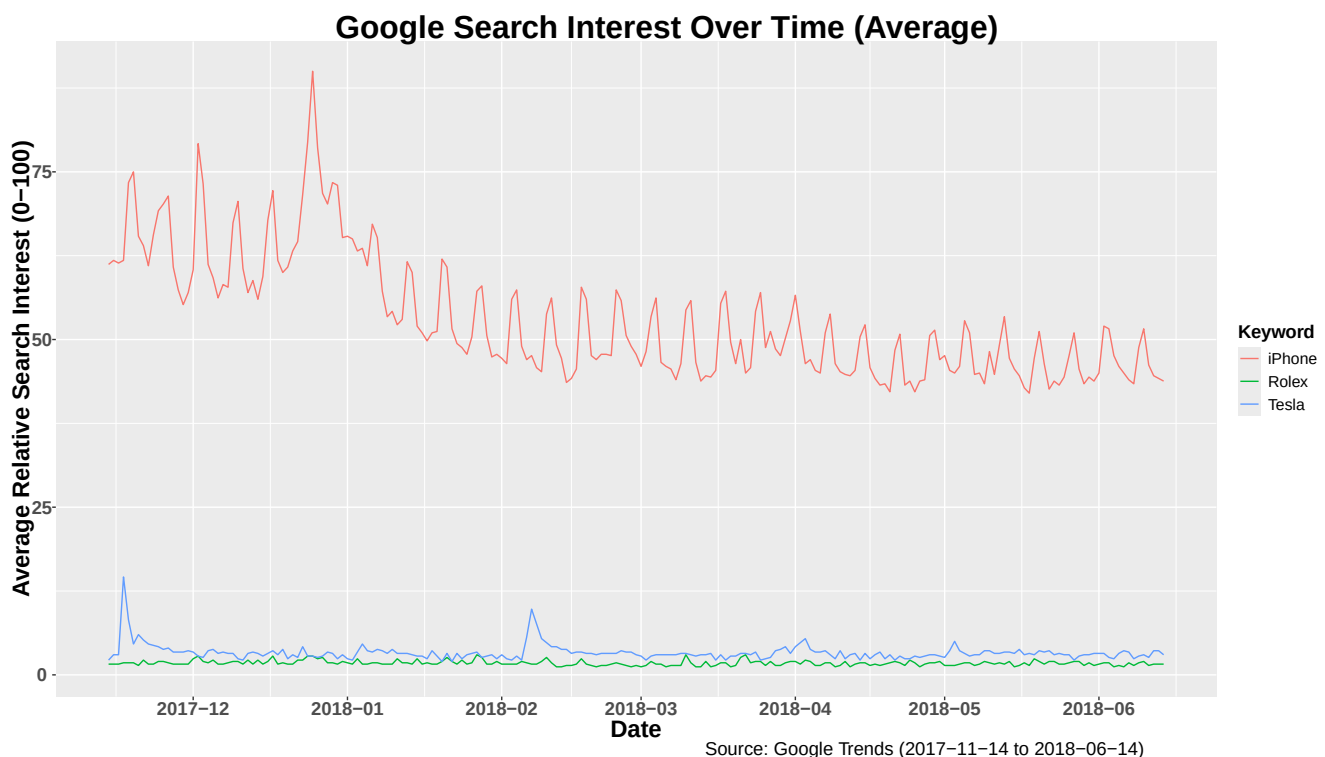
[9] "like_dislike_ratio" "comments_views_ratio" "likes_views_ratio" "dislikes_views_ratio" [13] "hits" "youtube_status"

3 Analysis and Visualisation

3.1 The Big Picture - Comparing the Brands

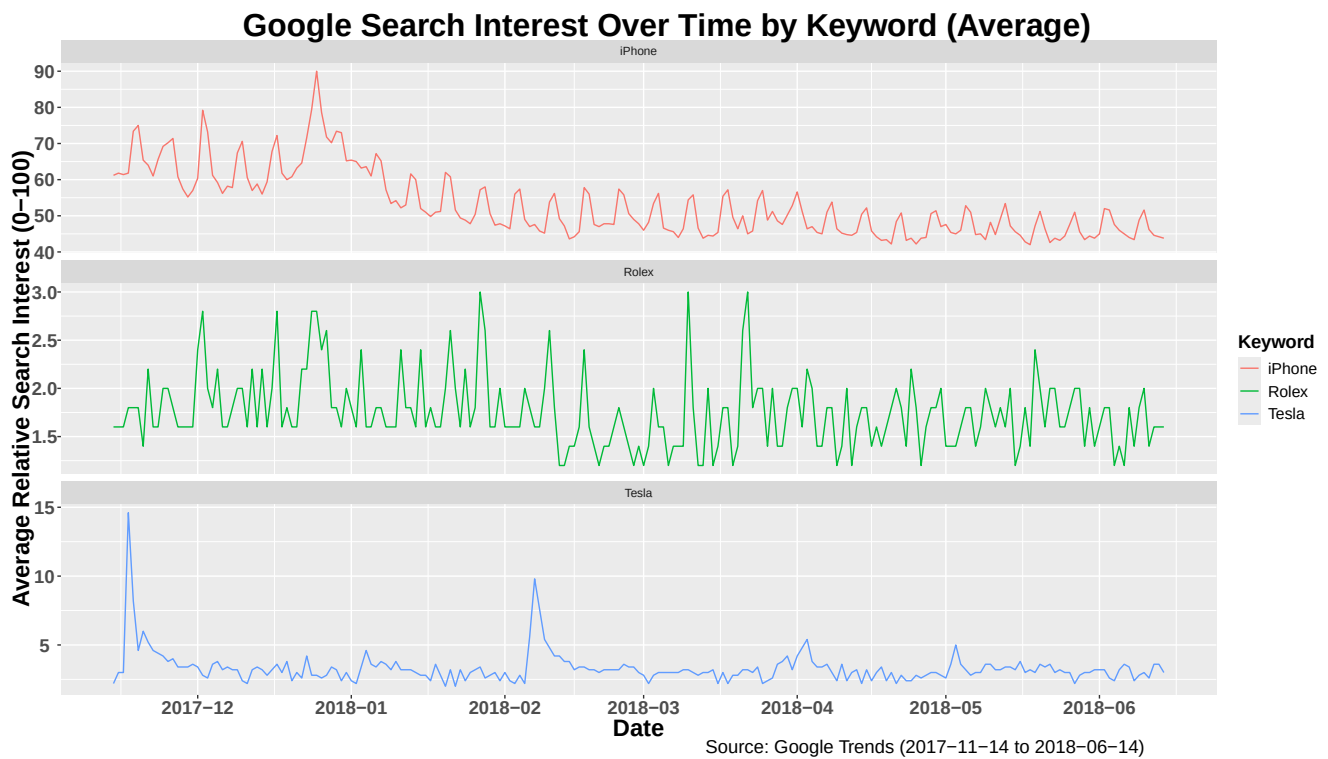
This chapter presents the **core** of our analytical project. First, We explore the datasets, **comparing** the brands on both platforms and provide a **big-picture overview**. Second, we focus on individual brands on Google platform, uncovering insights from different perspectives. Third, we investigate hidden patterns through **model analysis**, applying a linear regression model to the data. Fourth, we performed a YouTube-specific analysis of brand performance. Last but not least, by separating tags, we made **word clouds** to reveal what keywords were most associated with each brand, providing a new perspective for **product improvement** and marketing.

3.1.1 Graph 1: Overall Google Search Interest Comparison



To visualise **overall search interest** on Google for 3 brands, a graph with search interest data of 3 brands was made. In version 1, all brands are shown in one graph, and following this, v2 presents 3 brands in 3 separate graphs. Both graphs show the same result, but are suitable for different reading preferences.

Graph 1: Overall Search Interest Comparison (Facet Chart)v2



Interpretation of Google Search Interest Trends

Key observations include:

iPhone Dominates Search Interest:

“iPhone” consistently maintains the highest search volume, with significant peaks (e.g., near 100 relative interest) likely corresponding to **product launches** or major announcements (e.g., iPhone X in 2017, iPhone XS in 2018). It suggests sustained consumer curiosity and media hype around Apple.

Tesla shows moderate but steady interest:

“Tesla” exhibits **lower but steady** search interest, with occasional spikes. This aligns with Tesla’s brand history and vehicle releases (e.g., Model 3 rollout, massive quality scandals), and Elon Musk’s public visibility. We will also research the possible events that happened around that time later.

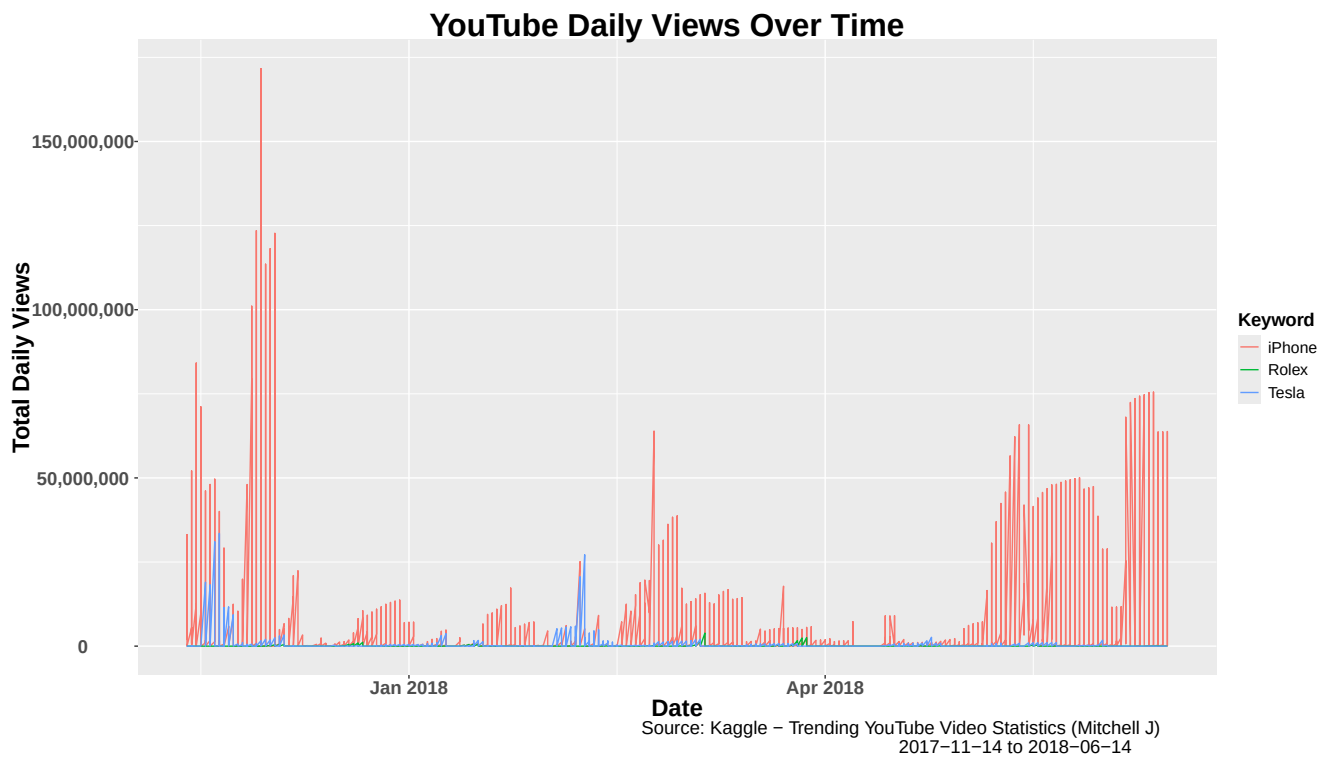
Rolex has negligible search presence:

The keyword “Rolex” shows near-zero search interest throughout the period. This indicates **lower public interest** compared to the other major brands. As a luxury watch brand, Rolex may not rely on hypes, otherwise, more marketing campaigns could be beneficial.

3.1.2 Graph 2: Overall YouTube Views Comparison

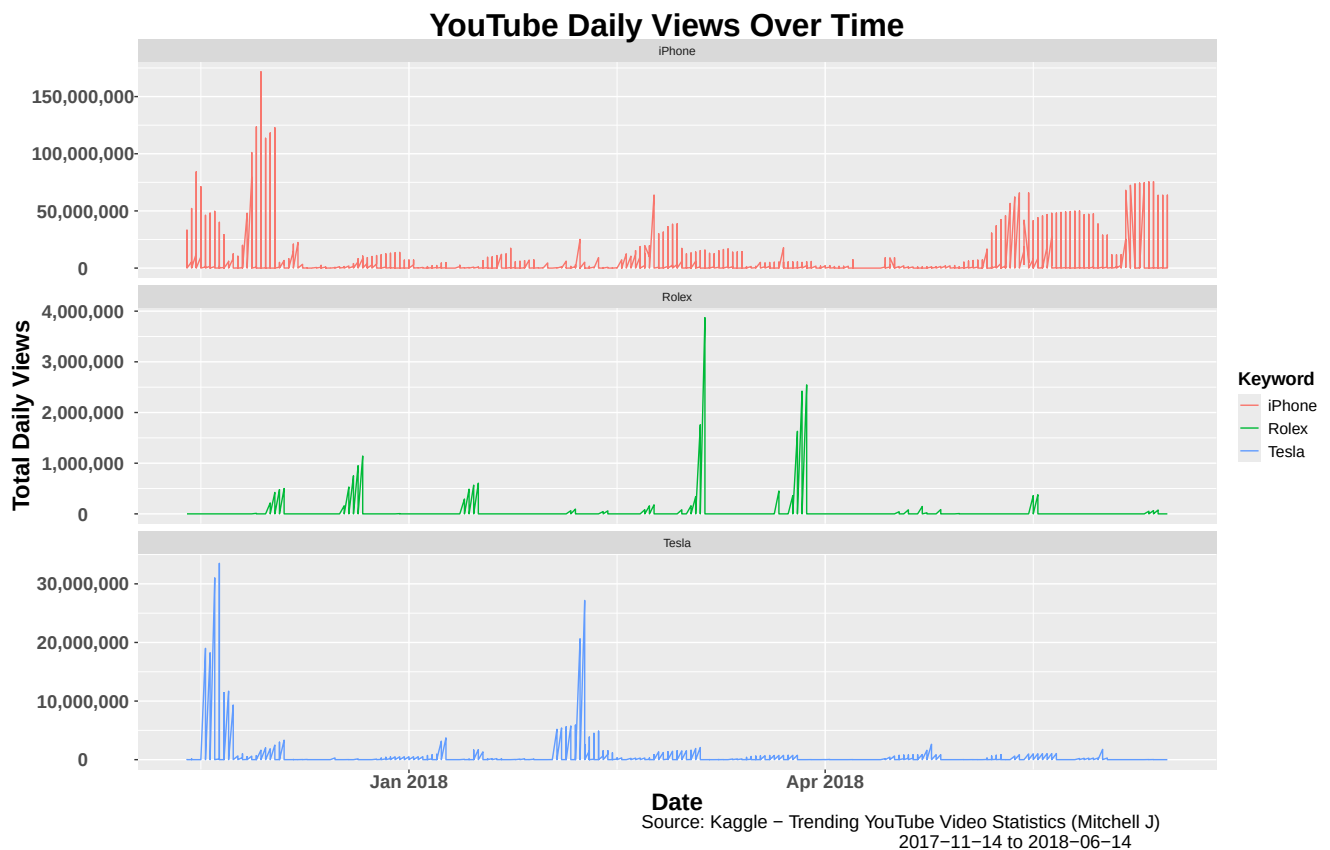
(Line Chart) v1

As with Google search interest, we did overall comparison for 3 brands on YouTube platform with 2 versions of graphs.



This graphs track the **daily view counts** for videos related to three keywords—iPhone, Roles, and Tesla, in 5 countries. The result indicates a massive disparity in viewership between iPhone and the other two keywords.

Graph 2: Overall YouTube Views Comparison v2 (Facet Version)



Interpretation of YouTube Daily Views Trends

iPhone: Viral and Event-Driven Engagement

The keyword “iPhone” shows an **astronomical level of views**, peaking at nearly 150 million daily views, which dwarfs the other terms. The trend is very **volatile**, with a dramatic spike in early March 2018 followed by a sharp decline.

This peak is hypothetically due to a **product launch** or major announcement. After researching, the timing suggests that it was driven by the release of the **iPhone X** (announced Sept. 2017 and launched on 3.Nov. 2017) or more likely. This pattern could suggest that consumers search for YouTube reviews, unboxings, tutorials, and software update related to new Apple products. And it lasted about one month.

Tesla: Consistent Niche Interest

Trend: The keyword “Tesla” shows a stable trend, ranging from 1 million to 3 million daily views. It shows **gentle fluctuations** but no sharp, viral spikes like the iPhone.

The steadier trend indicates a noticeable and continuous presence on the platform. However, certain events could influence the temporary interest of the public. For example, on 17. Nov 2017, Tesla announced its **semi truck** project. The futuristic design was followed by huge **media coverage**, which possibly cause a spike in views that lasted about 3 months. And in early 2018 there were huge discussions about Tesla falling short on Model 3 deliveries. In the same period, the stock price dropped approx. 33%. This indicates that both launch events and negative coverage could boost Tesla’s publicity on YouTube.

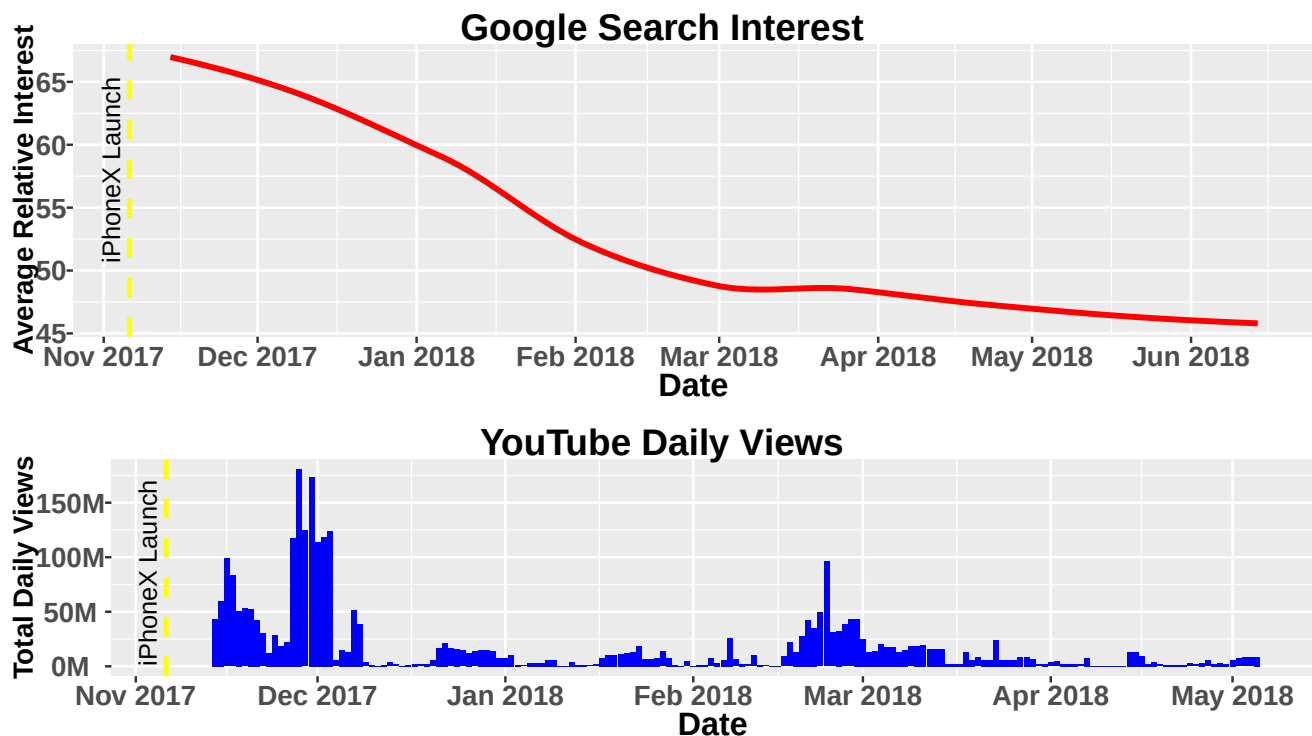
3.1.3 Graph 3: The iPhone Hype Cycle

This section of the big picture, we would like to have one graph that includes YouTube and Google data, and in the meantime, focuses on one product pattern, e.g., the Hype Cycle around product event like the iPhone X launch in Nov 2017.

To achieve this, 2 versions of the graphs are constructed, covering 180 days next to event. In v1, 2 sections of visualisation were drawn for YouTube and Google data. The Y axis represent YouTube and Google engagement and the X axis represents the time spans.

Graph 3 v1: The iPhone Hype Cycle (Google & YouTube separately)

180-day window around the iPhone X launch.



Source: Kaggle – Trending YouTube Video Statistics (Mitchell J) &

Google Trends (2018-01-01 to 2018-12-31)

Interpretation of Google Search Interest Trends

Google search interest hype lasted more than **3 months**, from November to February. Even though data before Nov is missing, we could see that starting from November, the search interest already had risen and peaked. It indicates that the launch event news had a great impact on iPhone Google search interest and the **elevated interest** lasted at least 3 months.

Google search strategy for client

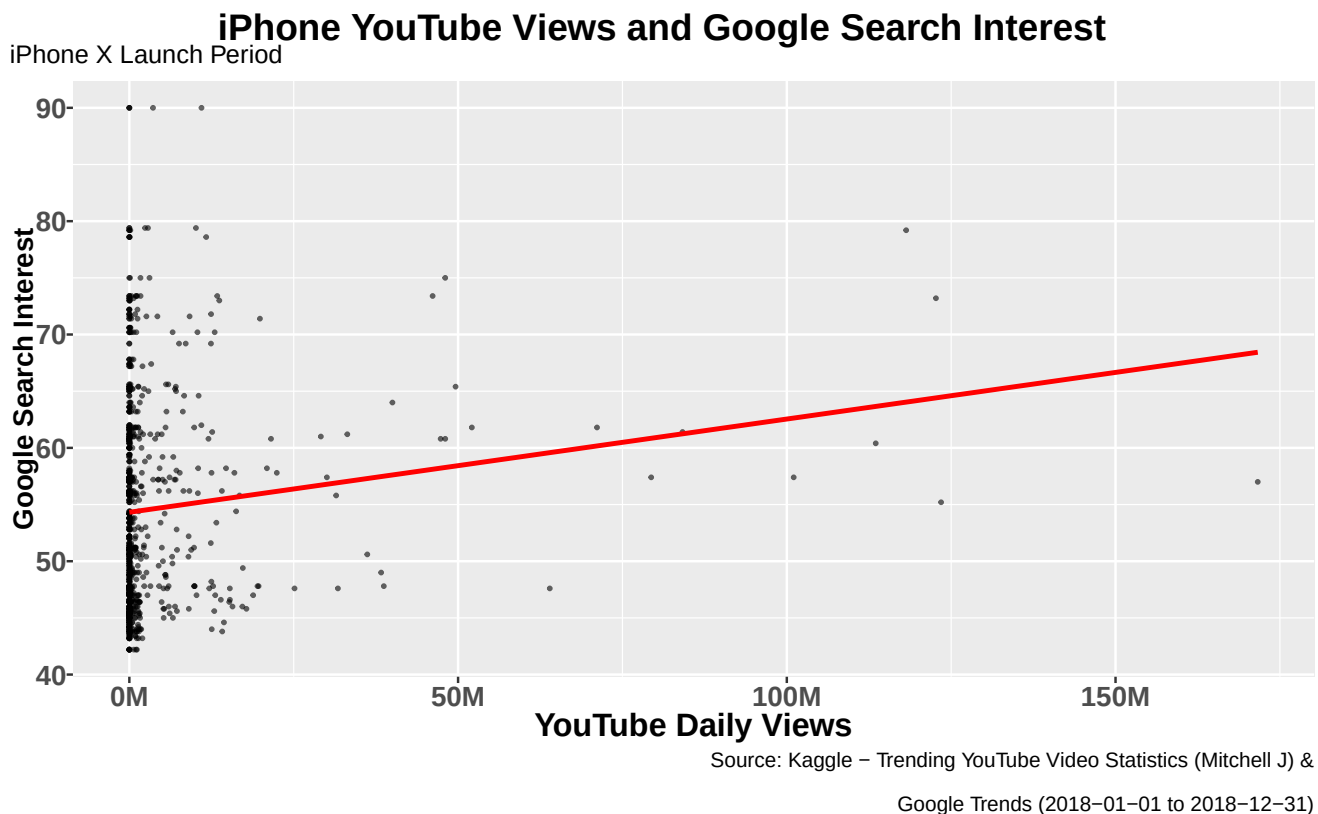
Apple can intensify Google platform **advertisement allocation** approx. 2 months before and 3 months after the major product launch event, to maximize customer interest and exposure on Google.

Interpretation of YouTube Views Trends

- Peak views in December: YouTube trending videos about iPhone started to emerge around Mid-November after the iPhone X launch. And the total views of these videos have peaked in December, with a daily total view reaching just over 175M per day. After that there is a sharp decline, with a small spike around March. The hype on YouTube is short lived and slightly scattered, only lasting about 1 months.
- YouTube strategy for client: Apple can start to focus on allocating marketing resources on YouTube **immediately** after the launch event of a major product, especially during the first month after the event. Continuous marketing effort on YouTube after that is **optional**.

Graph 3 v2: The iPhone Hype Cycle (Scatterplot)

180-day window around the iPhone X launch. To explore the big picture, using iPhone as an example, we plotted Google search interest on y axis and YouTube views on x axis. The red line represents the correlation of Google interest and YouTube views.



```
[1] "The correlation coefficient is: 0.116"
```

Interpretation of Google Search Interest Trends and YouTube total views

The scatterplots reveal dispersed data points, and the red line appears to show a **slight correlation**. However, is it really correlated as our **intuition** tells us? The result is 0.116. As we know, ± 0.0 to ± 0.3 equals very weak or no correlation.

We can safely say, even though **counterintuitive**, while there is a slight tendency for YouTube views and Google searches to move together, the relationship is **not strong** or predictable. High volumes of YouTube views on a given day were very weakly associated with high levels of Google search interest.

Platform Insight for client

The above suggests **user behavior** on these two platforms was largely **independent**, at least during the iPhone X launch period. It is beneficial for the clients to work with the 2 platforms with **independent approach** and strategy, for better result.

3.2 Google Search Trend Analysis

3.2.1 Loading and preparing the data

Because downloading Google Trends data is unreliable (more on this later in our *chapter of choice* about `gtrendsR`), we are working here with data that we previously have downloaded and saved as `csv` files. All five country files must be prepared in the same way, for which we have written a **function**. With `rbind`, we can easily concatenate the pre-prepared data and use `pivot_longer` to create a second **long** data set we will need for certain visualisations. With these steps we are now prepared for the following analyses.

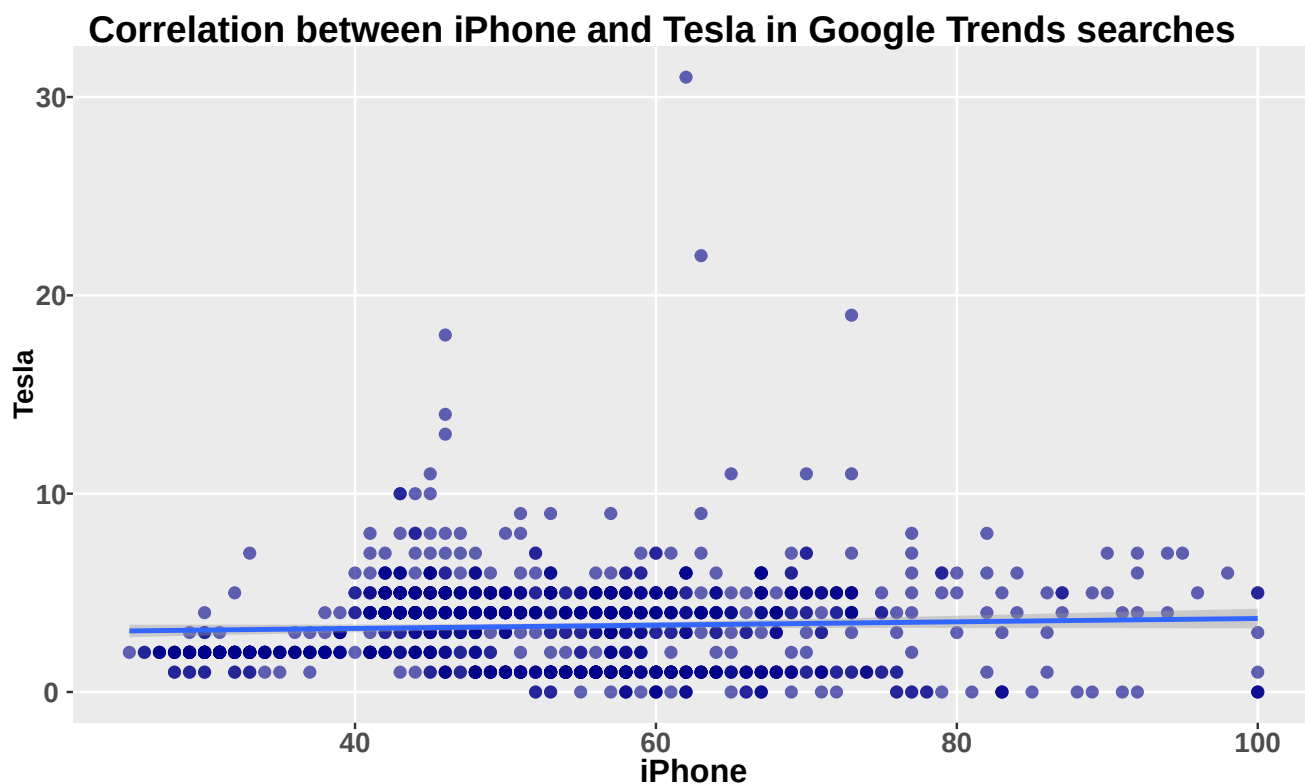
Data inspection: Besides `head()`, `View()`, `summary()` and `table()`, we used a **faceted line chart** to check whether the data for all countries and for the three keywords iPhone, Rolex, and Tesla has been imported correctly.

3.2.2 Correlations between brands

We examine if there is a correlation between **Google Searches for iPhone and Tesla**. Are days when iPhone searches are very prominent also days when Tesla searches account for a significant proportion? Or do searches for these two brands behave independently?

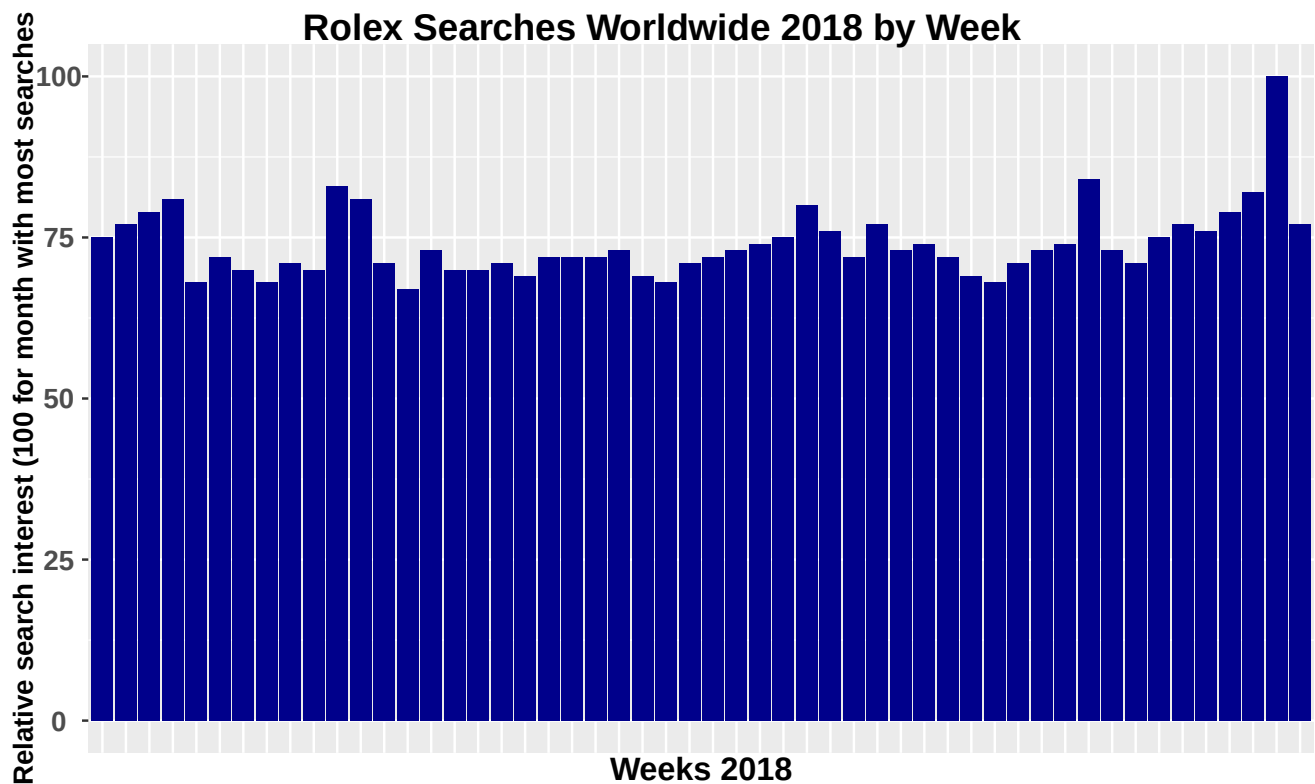
This question may be more relevant for other examples, such as when examining two **brands in the same industry**. Or when a brand is looking for a partner for **joint marketing activities**. In such cases, it can be very helpful to know whether such correlations exist.

Visually, it appears that Google searches for iPhone and Tesla are independent of each other. We believe this is an expected result.



3.2.3 Seasonality of Google searches

An important aspect of analyzing Google Trends data is **seasonality**. Are there periods when a particular brand is searched for more frequently? In the case of **Rolex**, for example, the annual **Baselworld** trade fair at the end of March or the **holiday season** at the end of the year could result in increased searches on Google.



Source: Google Trends (2018-01-01 to 2018-12-31)

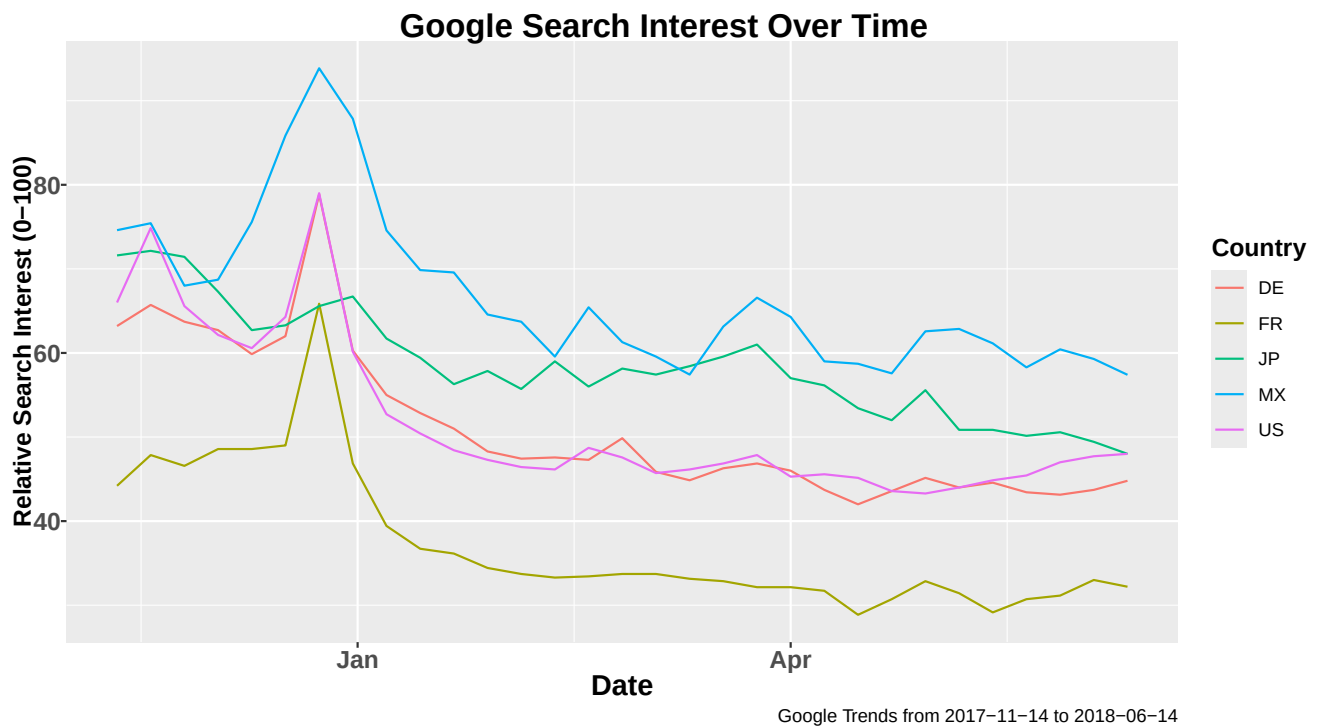
The weekly bar chart actually shows these **two peaks**. The first few weeks of the year, possibly as a echo from the festive season, show relatively strong search interest, followed by a first peak in **March** after a few weaker weeks. And in the penultimate week of the year, during the **holiday season**, the maximum is reached. Because no other brands relativize Rolex's values, Google Trend sets Rolex's maximum value at 100.

3.3 Model Analysis: fitting a linear regression model and visualize it

3.3.1 Data visualization for 5 countries

We want to find out how Google Searches for 'iPhone' in different countries are related. Are days with many searches for iPhone in the US also days with many searches in Germany? We would expect so, as iPhone is a global topic and news, e.g. the launch of a new iPhone version, swap across the globe in no time. But still, let's check if we can prove it with the data.

First, we do a line graph comparing the Google Trends data for iPhone country by country. To get a nicer graph, we aggregate the data from daily to weekly.



As we can see, the relative search interest in 'iPhone' is moving quite similar across all five countries. We now want to check with a model how closely two countries are related. We pick US and Germany, as we are most interested in these countries. As they are also behaving very similar over time, we expect a high correlation and a strong linear model explaining the interest in Germany together with the interest in the US.

3.3.2 Prepare the data for modelling and checking all correlations

As initial step, we have to prepare the data. We used the daily data as we want to have data as rich as possible for modelling. For the linear regression model `lm()` we need separate columns for iPhone search interest per country. As a first analysis, we calculate all correlations and visualize them with `corrplot`. In particular, the correlations between `iPhone_DE` and `iPhone_US` are very high, even close to 1 (0.93).

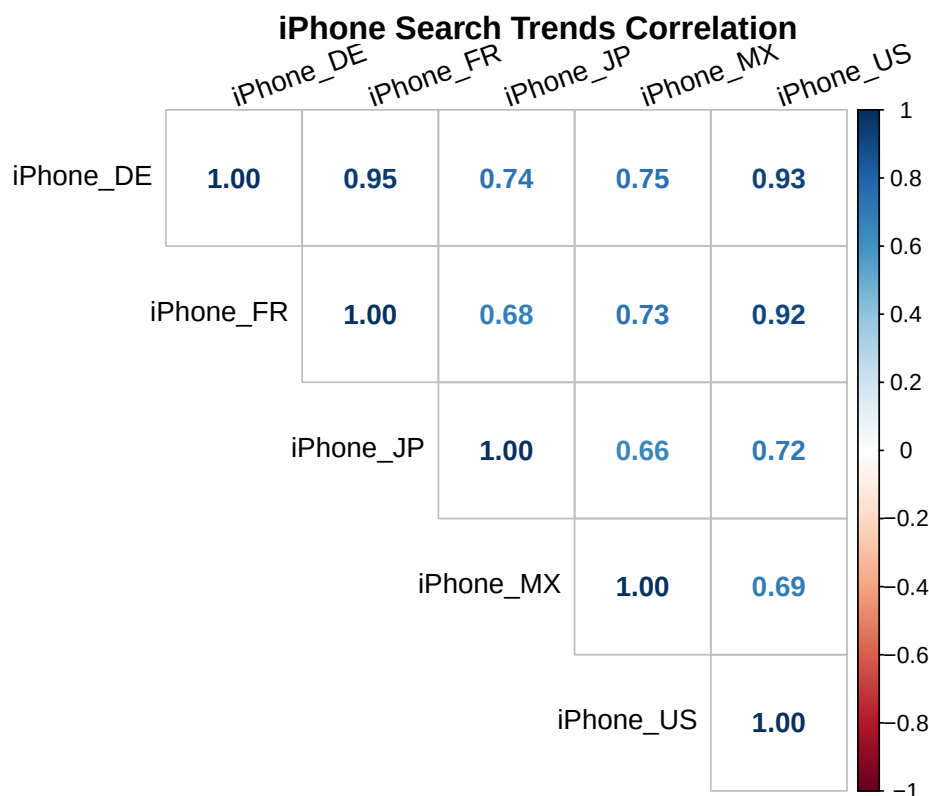


Figure 3.1: Source: Google Trends (2017-11-14 to 2018-06-14)

3.3.3 Fitting a linear model

With the data already prepared the fitting of a linear model is straight forward and with `summary()` we get all relevant information about the fit.

```
## Fit the linear regression model and show a summary
model <- lm(iPhone_DE ~ iPhone_US, data = lm_data)
summary(model)
```

Call:

```
lm(formula = iPhone_DE ~ iPhone_US, data = lm_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-15.6106	-2.2632	0.0063	2.1280	21.1719

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.59440	1.30825	4.276	2.88e-05 ***
iPhone_US	0.87825	0.02479	35.433	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.788 on 211 degrees of freedom

Multiple R-squared: 0.8561, Adjusted R-squared: 0.8554

F-statistic: 1255 on 1 and 211 DF, p-value: < 2.2e-16

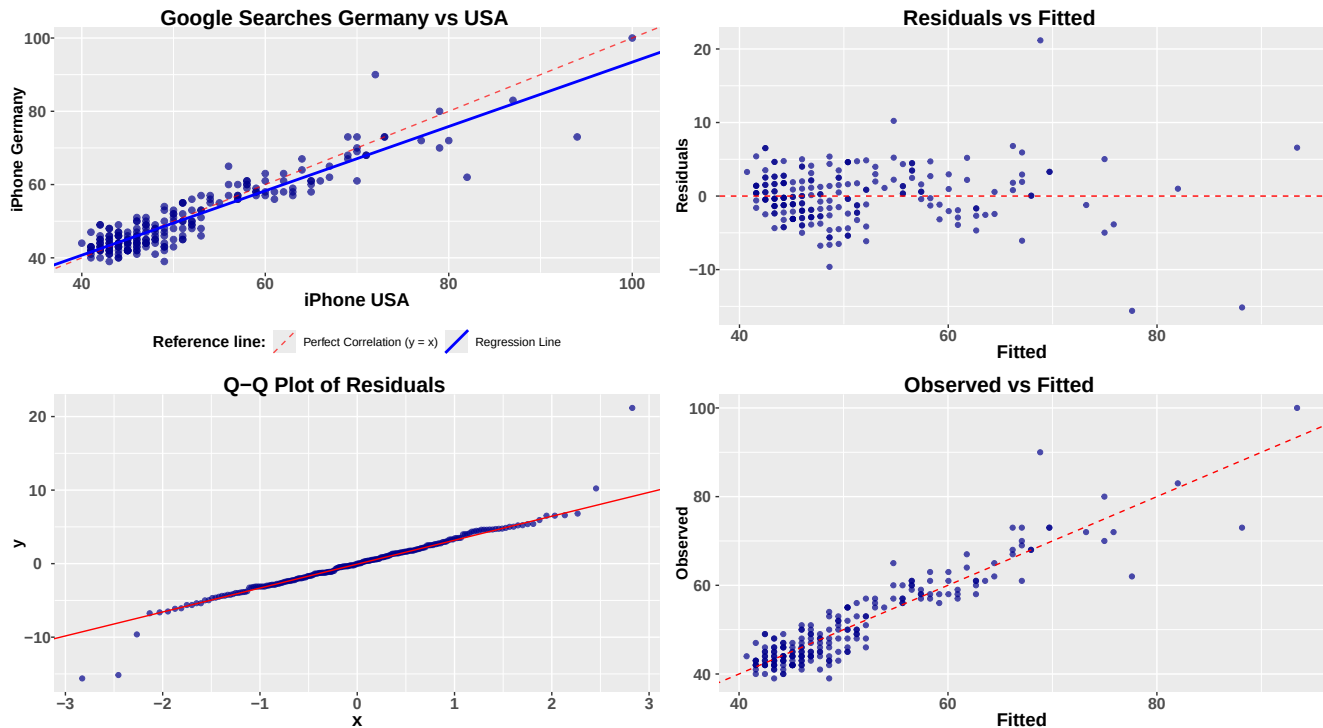
As expected the linear regression model shows a strong relation between the Google search interest in iPhone between the US and Germany: The p-value of the model is close to 0 and therefore significant. The same is true for the iPhone_US estimated coefficient of 0.878. This means an increase of interest by 1 in the US goes along with an increase of interest in Germany by 0.878.

3.3.4 Visualize the linear regression and the quality of the model

We create four visualizations for the model. First, a scatter plot showing all data points (i.e. days) with iPhone_US on the x and iPhone_DE on the y axis. We add a line for the perfect correlation ($y = x$) and the regression line calculated with the model. Thereafter, we extract intercept and slope from the coefficients that we received when fitting the model.

Next, we generate three standard graphs that are used to check the quality of a linear regression model:

- **Residuals vs. Fitted** to check linearity and constant variance: Any clear pattern indicates an issue.
- **Q-Q-plot of Residuals** to check the normal distribution of errors. Points should follow the line.
- **Observed vs. Fitted** to check visually how good model accuracy and predictive power are.



Source: Google Trends. Each point represents a day from 2017-11-14 to 2018-06-14.

To summarize, there are no major issues with the model. It explains the Google search interest in Germany very well based on the US Google search interest. Therefore, the result of the analysis confirms our initial expectation.

3.3.5 Conclusion for our clients

For any brand, it is important to know the level of interest (potential) clients show over time. In particular, when a major event is planned, monitoring of Google Search should be set up to understand how effectively the event triggered interest, how the interest developed afterwards and how it was in the markets/countries that the brand is most interested in.

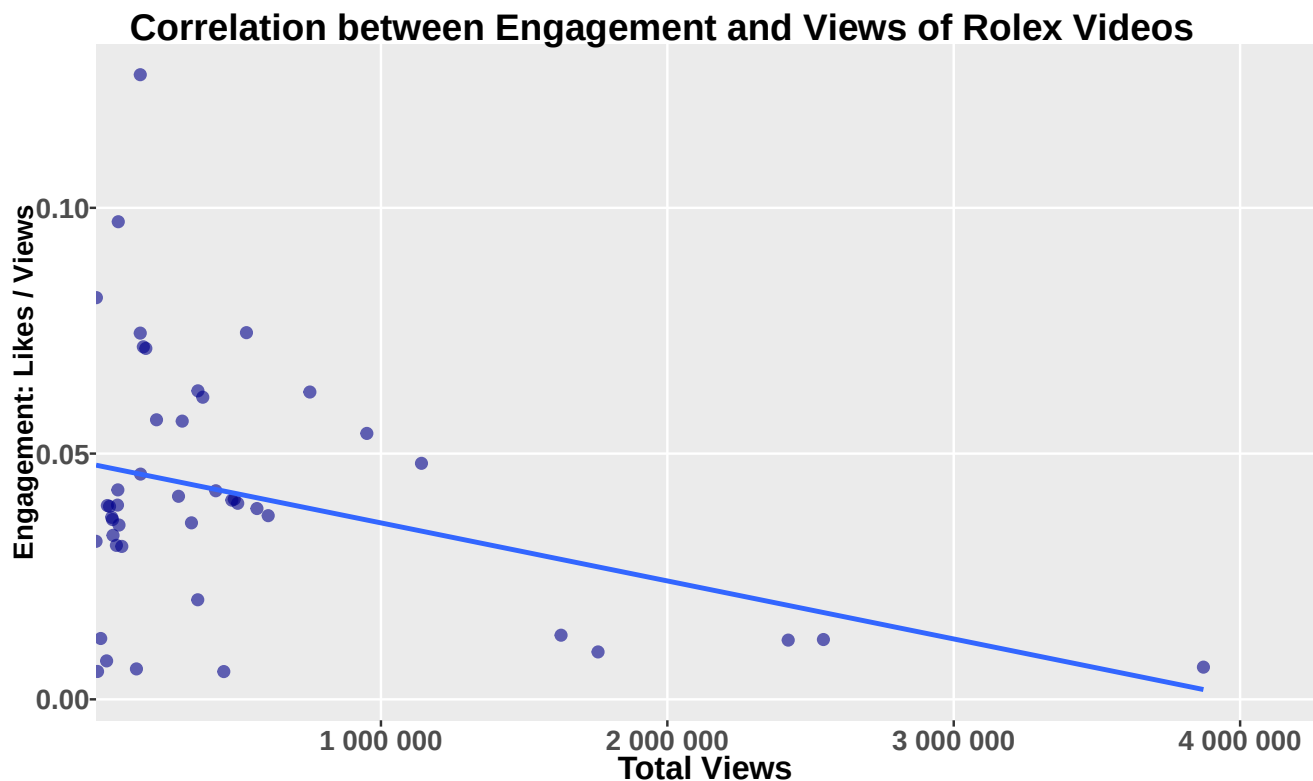
In some countries, the event may have caused a sharp increase in interest that will last for a long time. While in other countries the effect may be rather short-lived. It is essential for brands to be aware of such effects in order to respond appropriately with marketing measures.

3.4 YouTube Video Analysis

Now let's take a look at the YouTube data with **three graphs**.

3.4.1 Positive YouTube engagement

An important indicator is user **engagement**, which is measured by the number of likes per view: if a video is rated with a like, this indicates user engagement. We look at the relationship between engagement and total views and find that as videos become more popular, the number of likes per view decreases. This is understandable: very popular videos are also watched by people who are less interested in the topic. Therefore, the **popularity of the video** must always be taken into account when interpreting engagement.

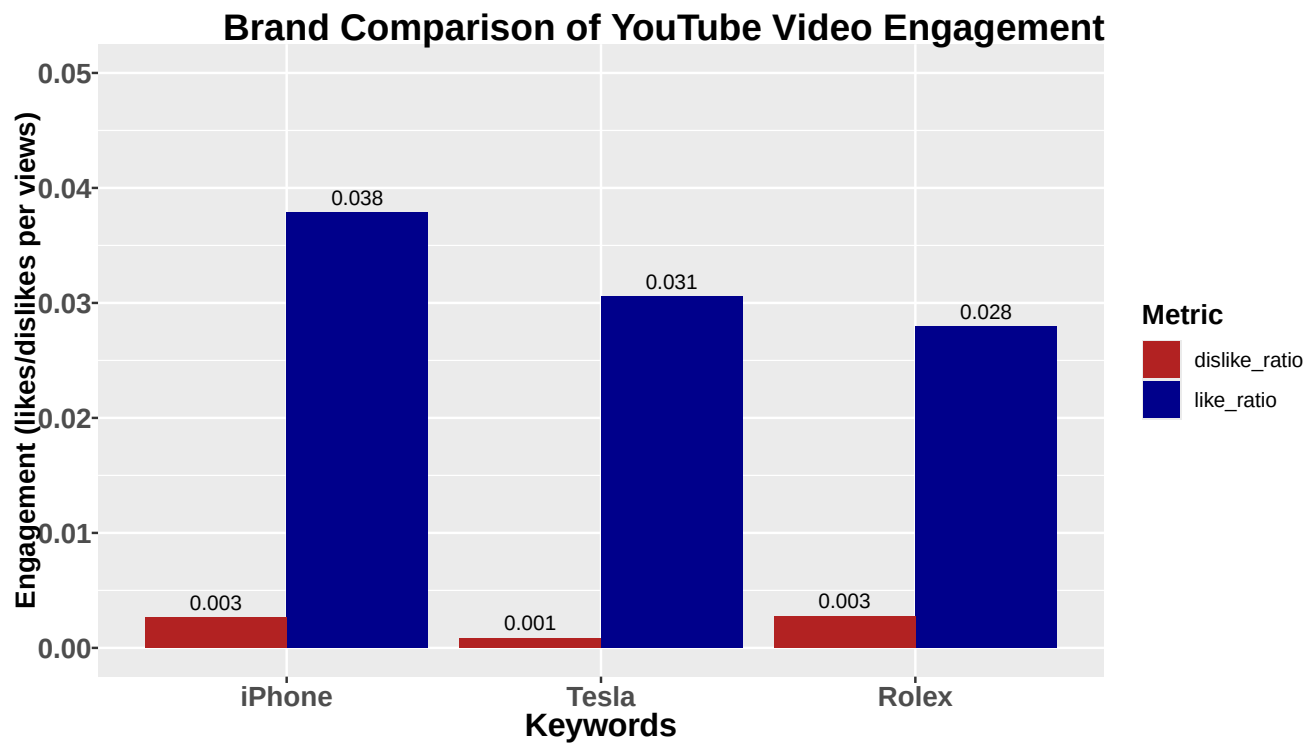


Source: Google Trends. Each point represents a day from 2017-11-14 to 2018-06-14.

3.4.2 Negative YouTube engagement

Engagement also exists **in a negative sense**, as the number of dislikes per view. Next, we are interested in the ratio of positive engagement (likes per view) to negative engagement (dislikes per view) for the videos of the three brands. We can see that the videos of all three brands receive **far more likes than dislikes**. In relative terms, however, it is striking that Tesla videos receive three times fewer dislikes (0.001) than the iPhone and Rolex videos (0.003).

For our **potential customers**, it is important to know the positive and negative engagement of the users of their videos, to track it over time, and to compare it with the videos of their competitors in order to be able to **estimate the corresponding brand effects**.



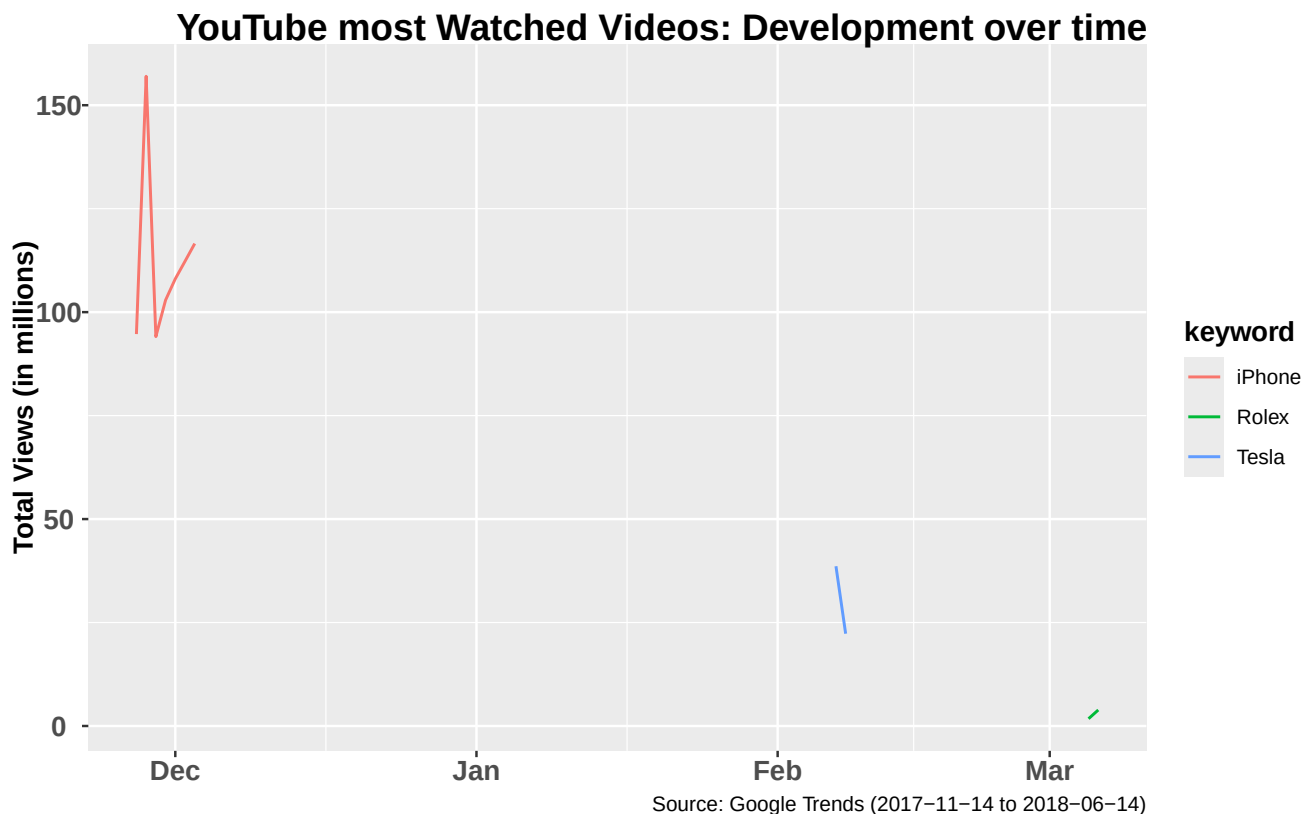
Source: Kaggle – Trending YouTube Video Statistics (Mitchell J) &

Google Trends from 2017-11-14 to 2018-06-14

3.4.3 Longevity of YouTube videos

We have already seen that an event has significantly less resonance in YouTube usage than in Google searches. That's why we now want to look at the **longevity** of videos and have selected the **most-watched videos** of the three brands for this purpose. We are interested in how their use develops over time.

The following graph shows an **unexpected problem**: the most-watched **videos disappear** from our data after a few days.



We have **two possible explanations**:

- Either the cryptic ID of a video, which we use to track the use of videos over time, changes after just a few days. This could be due to technical reasons how YouTube is setting these IDs.
- Or the data in the Kaggle dataset shows only a limited YouTube video universe.

In any case, it is completely **unrealistic** that a video that has been viewed millions of times would no longer be viewed from one day to the next.

In any case, we have decided to **investigate this phenomenon** at a later stage, as we are convinced that the longevity of videos can be of great interest to our potential customers.

3.5 WordCloud

After all these analyses and visualizations of essentially 'numbers', we should not forget that brands and branding are primarily about **content**. Here we have good news: YouTube and Google Searches are too! One can and should examine what the actual content of the videos on YouTube, besides the fact that they are tagged with the keywords 'Rolex' or 'iPhone'. With a wordcloud, we can give a very good first impression of the content based on all tags that were given to the videos. We believe that understanding what content of the videos is, is as important as knowing how often they have been watched.

First, we create a wordcloud for **Rolex**. This means: We are counting all tags given to all Rolex videos. As a reminder: A Rolex video is any video that contains "Rolex" in the title or in the tags. Then we select the most frequent tags (besides Rolex) and arrange them in a roughly circular layout. The most frequent tags/words are placed first in the center and with a larger font, while less frequent words get a smaller font and are placed later in available space around them.

The R package `wordcloud2` provides many parameters which allow us to customize looks of the wordcloud. Without going to much into detail **the process has two steps**:

1. Prepare the words to be displayed in a **dataframe with two columns**: the words and their counts. Regarding the words, some cleaning is needed, like removing white spaces and stopwords (words with no relevance for the wordcloud, e.g., 'and'). And when you see the first wordcloud you will most likely remove other words that you don't want to be shown or words that are just written differently that you want to combine.

2. Call the `wordcloud2()` function and define all **visual parameters**. This step requires a trial-and-error approach to set all size parameters correctly to get a nice wordcloud. The documentation of the package is good, so you know what each parameter does. But still you will be surprised how minor changes make the wordcloud look good or rather useless.

Now comes the preparation of the data. A short note: some packages loaded here are not necessary for the wordcloud but they allow more functionalities, e.g. check available fonts (`systemfonts`) or to export the wordcloud image as pdf (`webshot2` and `htmlwidgets`).

3.5.1 Rolex wordcloud

Then we create the Rolex wordcloud. We restrict the data to words with more than 100k views and set a specific palette that represents the corporate identity of Rolex. And we set the font to Garamond which is close to the font Rolex uses.

```
# Define luxurious colors that match the Rolex brand
pal_rolex <- c("#228B22", "#0A1931", "#B09770", "#722F37", "#B36A5E",
              "#454545", "#006D77", "#4C3F91")

# Create the Rolex wordcloud
wc_obj <- wordcloud2(
  rolex_tags_cleaned_unique %>% filter(total_views > 100000),
  size = 0.3, rotateRatio = 0,
  minRotation = -pi/2, maxRotation = pi/2,
  fontFamily = 'Garamond',
  fontWeight = 800,
  gridSize = 4, shuffle = TRUE,
  shape = 'circle', ellipticity = 0.5,
  color = rep_len(pal_rolex, nrow(rolex_tags_cleaned_unique)))

## Screenshot only for PDF render
if (knitr::is_latex_output()) {
  html_file <- "temp_wordcloud_rolex.html"
  htmlwidgets::saveWidget(wc_obj, html_file, selfcontained = TRUE)

  webshot2::webshot(
    html_file,
    "rolex_wordcloud.png",
    delay = 3,
    vwidth = 700,
    vheight = 500
  )

  knitr::include_graphics("rolex_wordcloud.png")
} else {
  wc_obj
}
```


4 Chapter of choice: The R Package *gtrendsR*

The package **gtrendsR** provides an R interface to the **Google Trends API**. It enables us to download and analyze data about how frequently certain terms are searched on Google. Search interest can be studied across different **regions**, **time windows**, and **categories**, which makes the package useful for projects that aim to investigate public attention toward specific topics.

4.1 Core Function

The core function of the package is: `gtrends()`, which sends queries to the Google Trends API and returns the result.

4.2 Main Parameters

Some of the most relevant arguments that we can specify in `gtrends()` are:

- **keyword**: One or more search terms (character vector).
- **geo**: Geographic scope of the search. Examples:
 - "US" → United States
 - "DE" → Germany
 - "" → worldwide
- **time**: The time frame of interest. Possible values include:
 - "now 7-d" → past 7 days
 - "today 12-m" → past 12 months
 - "YYYY-MM-DD YYYY-MM-DD" → custom date range
- **gprop**: The Google search property (some examples):
 - "web" (default)
 - "news"
 - "images"
- **category**: Restricts results to a specific category (e.g., "67" for Finance).

4.3 Structure of the Results

A typical call to `gtrends()` returns a **list of data frames**. The most commonly used components are:

- **interest_over_time**
Time series of relative search interest (scaled 0–100) for the chosen keyword(s).
- **interest_by_country**
Aggregated search interest at the country level.
- **interest_by_region**
Regional breakdown (e.g., states, cities, DMA) depending on `geo` parameter.
- **related_topics**
Topics that are often searched together with the given keyword(s).

4.4 Summary

In summary, the package **gtrendsR** is a powerful tool that allows us importing Google Trends data directly into R. With **gtrendsR**, we can:

- Track search interest **over time**,
- Compare search intensity **across regions**, and
- Explore **related queries and topics** that frequently appear alongside our keywords.

4.5 Interpreting Google Trend Data

The **values from 0 to 100** on Google Trends represent **normalized, relative search interest**, not an absolute number of search queries. The value 100 indicates the peak popularity of a search term in a selected time period and region. **All other values are relative to this peak value**. For example, a value of 0 means that there is insufficient data or that the search term has a very low search volume.

How the numbers work:

1. **Normalization**: Google Trends normalizes the data on a scale from 0 to 100 to represent search interest in relation to all search queries.
2. **Relative values**: The values are not absolute quantities, but show how often a term was searched for in comparison to its own peak.
3. The number 100: This represents the time or region when a search term reached its peak popularity.
4. The number 0: This means that there was insufficient data for the search term at that time or in that region.
5. **Example**: If a search term receives a 50, this means that search interest was only half as high as the peak value, which is 100.
6. **Keep in mind**: If you combine a search with a very popular and a rather rare search term, the rare search term will get (very) low numbers. But when you are only searching for the rare term it will also get a 100 once, namely for the one day when it was most often searched for.

4.6 Example: World map with Google Trends data for Tesla

4.6.1 Import libraries

4.6.2 Load data

Google is restricting the download-limit of Google Trends data. Sometimes already after one download, it is not possible to make another download. As the behavior is - also based on comments of other users - very unpredictable, we saved successful downloads as RData objects and load the data whenever needed.

4.6.3 Prepare the interest_by_country data in gtrends

The `interest_by_country` dataframe, which contains geographic search data, required two steps before it could be used in analysis.

First, We have to adjust some location iso-codes so they fit to the map data. Second, the main metric 'hits' contains '<1', which makes the column a character string. We replace <1 and NA with 0 and set the column to numeric.

```
country_data_Tesla <- res$interest_by_country %>%
  mutate(
    iso2c = countrycode(location, "country.name", "iso2c",
                        custom_match = c("Kosovo" = "XK")),
    iso3c = countrycode(location, "country.name", "iso3c",
                        custom_match = c("Kosovo" = "XKX")),
    hits = case_when(
      hits == "<1" ~ 0,
      is.na(hits) ~ 0,
```

```
TRUE ~ as.numeric(hits)
)) %>%
select(iso2c, iso3c, country = location, hits, keyword) %>%
arrange(desc(hits))
```

4.6.4 Load and prepare the map data using ne_countries function from the rnatrualearth package.

Some missings can be replaced with values from other columns (e.g. France with its overseas regions was a bit of an issue). Then the data is left joined with the prepared Google Trends data.

```
## Get world map data
world_sf <- ne_countries(scale = "medium", returnclass = "sf") %>%
  mutate(
    iso_a2 = if_else(iso_a2 == "" | is.na(iso_a2) | iso_a2 == "-99", iso_a2_eh, iso_a2),
    iso_a3 = if_else(iso_a3 == "" | is.na(iso_a3) | iso_a3 == "-99", iso_a3_eh, iso_a3),
    iso_n3 = if_else(iso_n3 == "" | is.na(iso_n3) | iso_n3 == "-99", iso_n3_eh, iso_n3))

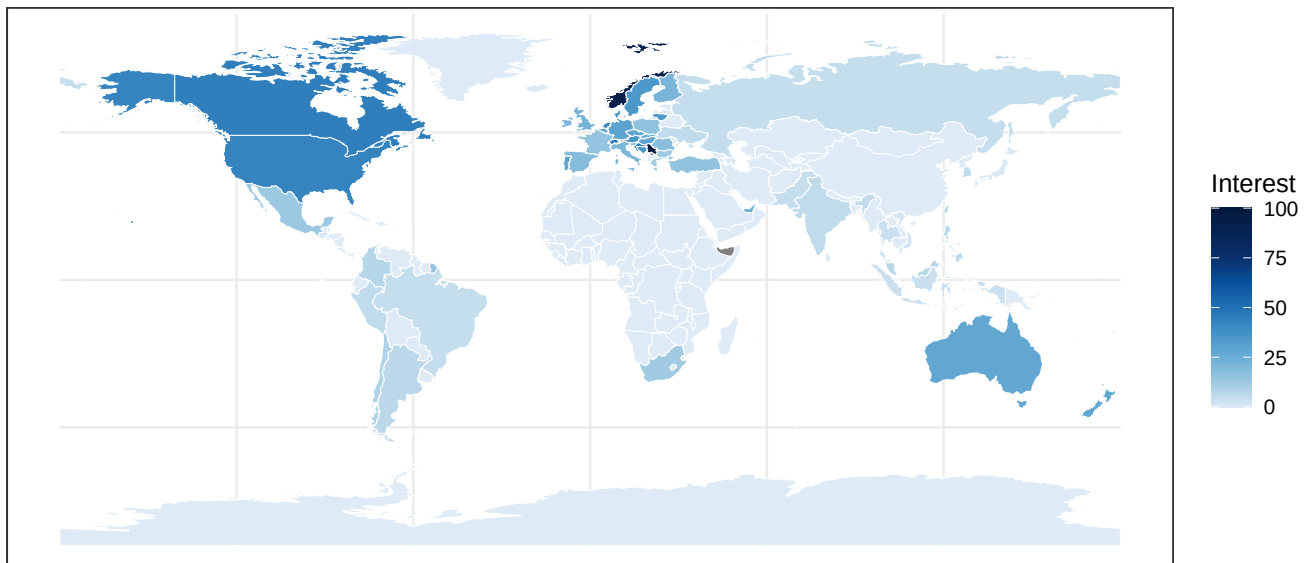
## Join by ISO3
world_trends <- world_sf %>%
  left_join(country_data_Tesla, by = c("iso_a3" = "iso3c"))
```

4.6.5 Producing the world map with geom_sf

Additionally we specify a color coding from bright to dark blues. The map shows the highest interest in Norway, where electric cars appear to enjoy significant tax advantages.

Google Trends: Interest in Tesla

Average interest level



Source: Google Trends (2017-11-14 to 2018-06-14)

5 Generative AI Usage and Guidelines

In this project, Generative AI was used as a supplementary tool on top of the assisting material that was provided in R-Bootcamp, for example, help function in Rstudio, vignettes in CRAN. In addition, we already had some script and Quarto templates from previous projects. All these sources formed the basis of our work with YouTube and Google data. :

In order to improve efficiency but also enhance our learning, the deployment of Generative AI is guided by a clear set of guidelines and strategies:

5.1 Guidelines for Responsible and Effective Usage of Gen-AI

1. Human-in-the-loop verification: for all Gen-AI content suggested by AI were to be seen as drafts instead of final result. Text, code or sources that were provided by AI needed to be critically reviewed, tested.
2. Emphasis on learning: Gen-AI should be treated as a learning tool instead of code generator or otherwise. After solutions being provided, students need to engage in critical thinking and ask follow-up questions to understand the logic and method.
3. Transparency: The use of Gen-AI should be openly acknowledged, especially where and how AI was used in the workflow. Only with transparency, the benefits of employing AI could be better understood and the quality of the project could be ensured.

5.2 Use cases of AI tools

We used AI-based assistance systems in specific cases. These primarily served to:

1. Brainstorming: Gen AI was used for elaborations of initial ideas, structuring thoughts and coding solutions.
2. Debugging and optimization: Gen-AI is a great tool for explaining error messages in understandable language. It is also suitable for optimizing non-functioning scripts that we had developed ourselves, which improves our project efficiency.
3. Proofreading: Gen-AI is deployed in accelerating the proofreading process, especially in detecting grammar and typo mistakes.
4. Information acquisition: Gen-AI is used for search of information about methodologies and code documentations, similar to how Google was used for searches. However, human verification of correctness and credibility of sources are crucial for us, because AI as a new technology that can provide inconsistent and incorrect results.

5.3 Benefits and challenges in using Generative AI tools

1. The usage of Gen-AI provides various benefits. For instance, when encountering errors in complex ggplot2 visualizations, Gen-AI immensely accelerated the debugging process. It could provide root cause of the error in minutes, which would normally take us 20 mins to search for answer on Stack Overflow or documentations. Moreover, Gen-AI provides excellent live-tutoring experience for students, especially for complex personal learning questions, which accelerates the learning process. In summary, Gen-AI usage made it possible for us to spend more time on analytical tasks and less on routine coding obstacles.
2. The challenges of using Gen-AI is also critically discussed. As Gen-AI usage can seem so easy for providing answers, there is the temptation to blindly trust the result in coding or information sources. This is where the guideline of always “Human-in-the-loop” verification comes into place again. We recognize the importance of us taking full ownership in our project, not AI.

6 Conclusion, limitations and outlook

6.1 Conclusions

6.1.1 Key Takeaways for Brands

- **Market Positioning:** Our **proof of concept** demonstrates the rich insights from Google and YouTube data for potential customers. It became clear that different brands play in very **different leagues**, regardless of their industry. Based on our analyses, we would describe the iPhone as a **global player**, Tesla as an **international player**, and Rolex as a **niche player** (although with a luxurious/international reach).
- **Marketing Resource Allocation:** YouTube engagement around the iPhone has massive spikes after **product launches** but fading quickly, while Tesla maintains interest. Compared with Google, where search interest lasts several months, YouTube hype is generally short-lived. This suggests our clients allocate **YouTube marketing immediately** after launches to catch **momentum**, while deploying **Google ads** over time to **maintain** popularity.
- **Predictive Insights & Seasonality:** We identified **correlations** between different brands and created a **linear model** that correlates search behavior in one country with behavior in another. Both approaches become even more important when examining brands in the same industry or the response to multinational advertising campaigns. And we were able to demonstrate the **seasonality** of search behavior on Google using the example of Rolex. This insight is of **direct benefit** to brand managers, as it helps them better manage slumps or extend peaks through campaigns.
- **Importance of Engagement:** Using YouTube videos, we were able to show that brand managers should keep an eye on **user engagement**, both positive in the form of likes/views and negative in the form of dislikes/views, and compare it with that of their competitors.
- **Content & Brand Alignment:** Finally, the word clouds reminded us that despite all the numbers and ratings, **content** should not be forgotten. The numbers for Rolex YouTube videos may look good. But is it also in the interest of brand managers if the primary focus is on showcasing German hip-hop artists?

6.1.2 Key takeaways for R-Bootcamp Learning

We implemented many different concepts and practices that we learned in the R bootcamp and also explored **new additional topics**. For example, we learned how to access current Google search data and how to analyze the content of YouTube videos using word clouds. By solving problems using the usual R tools, Google searches, and AI chatbots, we have now built **confidence** that we can tackle future data science challenges.

6.2 Limitations

Many limitations ultimately proved to be solvable problems that we were able to fix with a little data wrangling, for example. However, some aspects **fundamentally limited** our evaluations:

- **Outdated YouTube Data:** We worked with a Kaggle dataset **from 2017 to 2018**. This was sufficient for our purposes. In addition to this use case, which focuses on what can be gained from YouTube data, we would of course also like to investigate specific cases. To do this, access to current data is essential.
- **Changing Video IDs:** A second limitation arose when we attempted to track certain videos over a longer period of time to see how long they remained available. Apparently, the **cryptic video IDs change** from time to time, so we were unable to perform any meaningful analysis in this regard.
- **Google Trends API Restrictions:** We worked with data that we were able to download daily from Google Search Trends. One initial difficulty is that **Google restricts downloads** in a rather unpredictable manner. This significantly limits the workflow, as all downloads must be cached locally for security reasons.
- **Relative Google Trends Data:** However, the bigger problem with Google Search is that **only relative numbers** are available. It is not possible to obtain the absolute number of searches per keyword. All values are calibrated by Google Search Trends to 0-100 for each search query. On the one hand, this means **a significant loss of information** and makes it **impossible to compare different search queries** with each other.

- **Project Scope:** A final limiting factor was our somewhat **theoretical experimental setup**. With a more concrete case, our analysis would have been much more concise and the overall picture more coherent. The fact that we had data from 2017/2018 and were working with such data for the first time prompted us to examine somewhat more general questions, which we considered appropriate. But let's imagine Guinness Beers was our client and we could observe the launch of their first **non-alcoholic** Guinness Stout beer! That would, of course, give the analysis and the story a whole different dynamic!

6.3 Outlook

The limitations mentioned above lead to the main points we would address next:

- **Update to current data:** It is possible to **load current YouTube data**. However, this requires a Google Cloud project and an API key. Even then, there are certain limitations, but the data would definitely be more comprehensive than the data available to us in the `Kaggle` dataset. In this context, we would look at the `Tuber` package (CRAN), which can be used to load data on specific videos. For example, if Rolex decides to make its own videos available on YouTube to influence their brand image in its favor.
- **Explore Google Search Metrics:** We would also try to find a way to obtain **uncalibrated Google search data**. While initial attempts were proven challenging as Google intentionally limits our access to this data, the `Google Ads Keyword Planner` could be an alternative for future attempts.
- **Advanced Visualization:** A final area of improvement is regarding a small problem for the word clouds. It would have been nice to create the iPhone word cloud in the **format of the Apple logo**. However, even after an intensive search, we were unable to do this. We will not give up and will make a few more attempts.